

## KUR, Russian Federation

WMO: 316320

Lat: 49.933N

Lon: 134.633E

Elev: 928

StdP: 14.21

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -34.8      | -31.2 | -39.6                         | 0.6 | -34.4 | -35.9 | 0.7 | -30.5 | 14.5                  | 4.8  | 10.4 | 8.7  | 0.3                   | 20   | 0.565 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.4                   | 85.6            | 69.0 | 81.7 | 67.5 | 78.5 | 65.6 | 71.9                | 80.4 | 70.0 | 78.1 | 68.3 | 75.4 | 3.7                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.3                            | 111.6 | 75.5 | 67.4 | 104.3 | 73.6 | 65.8 | 98.4 | 71.6 | 36.5          | 80.7 | 34.7 | 77.9 | 33.2 | 75.6 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| (4) 10.4          | 8.2  | 7.0 | -39.5                      | 91.7 | 3.7                | 3.5 | -42.2                                              | 94.2 | -44.4      | 96.2 | -46.4      | 98.2 | -49.2      | 100.7 |    |    |
| (5)               |      |     | -39.7                      | 76.2 | 3.7                | 3.1 | -42.4                                              | 78.5 | -44.5      | 80.3 | -46.6      | 82.0 | -49.3      | 84.2  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 31.6   | -9.3  | 0.0   | 17.8  | 35.4  | 50.3  | 61.0  | 67.5  | 64.2  | 52.8  | 35.2  | 10.3  | -8.3  |
|                                                                        | DBStd       | 28.56  | 9.62  | 9.36  | 9.42  | 7.16  | 7.22  | 6.16  | 4.84  | 5.14  | 6.79  | 7.93  | 12.17 | 10.32 |
|                                                                        | HDD50       | 8276   | 1839  | 1401  | 999   | 441   | 87    | 2     | 0     | 1     | 48    | 462   | 1190  | 1806  |
|                                                                        | HDD65       | 12391  | 2304  | 1821  | 1464  | 888   | 457   | 150   | 29    | 74    | 368   | 925   | 1640  | 2271  |
|                                                                        | CDD50       | 1549   | 0     | 0     | 0     | 3     | 96    | 332   | 541   | 442   | 132   | 3     | 0     | 0     |
|                                                                        | CDD65       | 188    | 0     | 0     | 0     | 0     | 1     | 30    | 105   | 50    | 2     | 0     | 0     | 0     |
|                                                                        | CDH74       | 1972   | 0     | 0     | 0     | 2     | 107   | 484   | 914   | 426   | 39    | 0     | 0     | 0     |
|                                                                        | CDH80       | 524    | 0     | 0     | 0     | 0     | 18    | 139   | 284   | 79    | 4     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 2.1    | 1.4   | 2.1   | 2.6   | 2.6   | 2.5   | 2.4   | 2.2   | 2.0   | 2.0   | 2.1   | 1.9   | 1.4   |
|                                                                        | PrecAvg     | 27.7   | 0.4   | 0.4   | 0.6   | 1.4   | 2.8   | 4.1   | 5.9   | 5.3   | 2.8   | 1.7   | 0.9   | 0.6   |
| (15) Precipitation                                                     | PrecMax     | 38.8   | 1.1   | 1.5   | 1.8   | 4.8   | 4.7   | 9.4   | 13.3  | 9.7   | 6.0   | 3.5   | 2.5   | 1.8   |
|                                                                        | PrecMin     | 21.2   | 0.0   | 0.0   | 0.0   | 0.2   | 1.3   | 0.2   | 0.9   | 1.3   | 0.4   | 0.2   | 0.1   | 0.1   |
|                                                                        | PrecStd     | 5.2    | 0.3   | 0.4   | 0.5   | 1.1   | 1.1   | 2.2   | 2.8   | 2.0   | 1.5   | 0.9   | 0.7   | 0.4   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 22.2  | 30.7  | 48.4  | 71.0  | 82.0  | 89.6  | 92.2  | 86.2  | 78.3  | 65.4  | 41.1  | 27.1  |
|                                                                        |             | MCWB   | 19.7  | 24.4  | 37.5  | 51.4  | 59.0  | 70.0  | 71.4  | 70.9  | 64.5  | 51.3  | 34.1  | 23.5  |
|                                                                        | 2%          | DB     | 16.2  | 25.3  | 41.9  | 61.5  | 76.8  | 83.8  | 87.0  | 82.0  | 73.6  | 57.3  | 35.6  | 18.4  |
|                                                                        |             | MCWB   | 13.9  | 20.5  | 32.7  | 45.9  | 57.5  | 67.0  | 70.4  | 69.2  | 61.1  | 46.7  | 31.3  | 16.5  |
|                                                                        | 5%          | DB     | 11.0  | 20.6  | 37.6  | 55.1  | 71.7  | 79.4  | 82.7  | 78.5  | 69.8  | 52.3  | 32.4  | 12.8  |
|                                                                        |             | MCWB   | 9.4   | 17.2  | 30.5  | 42.1  | 54.5  | 64.1  | 69.3  | 66.8  | 59.0  | 43.6  | 29.5  | 11.5  |
|                                                                        | 10%         | DB     | 6.2   | 16.5  | 33.7  | 49.8  | 66.8  | 75.2  | 79.0  | 75.5  | 65.8  | 48.0  | 28.9  | 8.3   |
|                                                                        |             | MCWB   | 4.9   | 13.7  | 28.0  | 39.2  | 52.9  | 62.5  | 67.7  | 66.0  | 57.2  | 41.3  | 26.1  | 7.1   |
|                                                                        | 0.4%        | WB     | 19.8  | 24.9  | 37.8  | 52.5  | 62.7  | 75.1  | 75.1  | 74.1  | 67.2  | 53.3  | 35.0  | 23.7  |
|                                                                        |             | MCDB   | 22.1  | 30.5  | 47.3  | 68.8  | 78.0  | 84.3  | 85.9  | 82.0  | 75.9  | 60.4  | 39.2  | 27.5  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 14.1  | 20.8  | 33.3  | 46.8  | 59.0  | 69.3  | 72.4  | 71.2  | 63.6  | 48.6  | 32.3  | 16.6  |
|                                                                        |             | MCDB   | 16.1  | 25.0  | 40.1  | 59.3  | 71.6  | 78.6  | 81.4  | 78.4  | 69.8  | 55.9  | 34.8  | 18.4  |
|                                                                        | 5%          | WB     | 9.3   | 17.3  | 30.9  | 43.1  | 56.5  | 66.4  | 70.6  | 69.2  | 60.9  | 45.1  | 29.6  | 11.4  |
|                                                                        |             | MCDB   | 10.9  | 20.8  | 37.0  | 53.3  | 68.7  | 76.1  | 79.1  | 75.7  | 66.7  | 50.4  | 32.5  | 12.7  |
|                                                                        | 10%         | WB     | 5.0   | 13.7  | 28.3  | 39.9  | 53.8  | 64.0  | 69.1  | 67.3  | 58.5  | 41.9  | 26.1  | 7.0   |
|                                                                        |             | MCDB   | 6.2   | 16.3  | 33.5  | 48.6  | 64.2  | 72.6  | 77.0  | 73.1  | 64.4  | 47.1  | 28.8  | 8.3   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 25.9  | 29.3  | 26.3  | 22.6  | 24.5  | 22.6  | 19.4  | 19.7  | 22.7  | 21.4  | 22.5  | 23.1  |
|                                                                        |             | MCDBR  | 25.4  | 31.7  | 28.0  | 33.8  | 35.7  | 31.9  | 28.2  | 26.5  | 30.2  | 28.5  | 21.5  | 20.6  |
|                                                                        |             | MCWBR  | 23.3  | 27.4  | 19.7  | 19.8  | 17.9  | 16.0  | 13.6  | 14.4  | 17.5  | 17.5  | 16.6  | 18.9  |
|                                                                        | 5% WB       | MCDBR  | 25.1  | 31.1  | 26.2  | 30.9  | 30.2  | 26.0  | 22.7  | 21.5  | 24.3  | 26.3  | 19.9  | 20.0  |
|                                                                        |             | MCWBR  | 23.0  | 26.6  | 18.4  | 18.8  | 16.8  | 14.7  | 12.4  | 13.0  | 15.3  | 17.4  | 15.3  | 18.4  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.273 | 0.314 | 0.389 | 0.517 | 0.570 | 0.510 | 0.501 | 0.431 | 0.392 | 0.403 | 0.335 | 0.278 |
|                                                                        | taud        |        | 2.125 | 2.010 | 1.855 | 1.722 | 1.740 | 1.990 | 2.067 | 2.273 | 2.295 | 2.068 | 2.067 | 2.105 |
|                                                                        | Ebn at Noon |        | 238   | 252   | 250   | 229   | 224   | 241   | 241   | 255   | 252   | 216   | 204   | 213   |
|                                                                        | Edh at Noon |        | 25    | 37    | 52    | 67    | 69    | 54    | 49    | 38    | 34    | 35    | 27    | 23    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 536   | 922   | 1393  | 1538  | 1599  | 1718  | 1574  | 1422  | 1148  | 767   | 505   | 413   |
|                                                                        | RadStd      |        | 16    | 27    | 79    | 160   | 153   | 215   | 153   | 119   | 106   | 56    | 36    | 22    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (11 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page